

Experiment # 05

Implementation of Bridges and Spanning Tree Protocol

Objective:

The objective of this lab is to reinforce the concepts related to bridges and spanning tree protocol.

Software Tools:

- Cisco Packet Tracer

Theory:

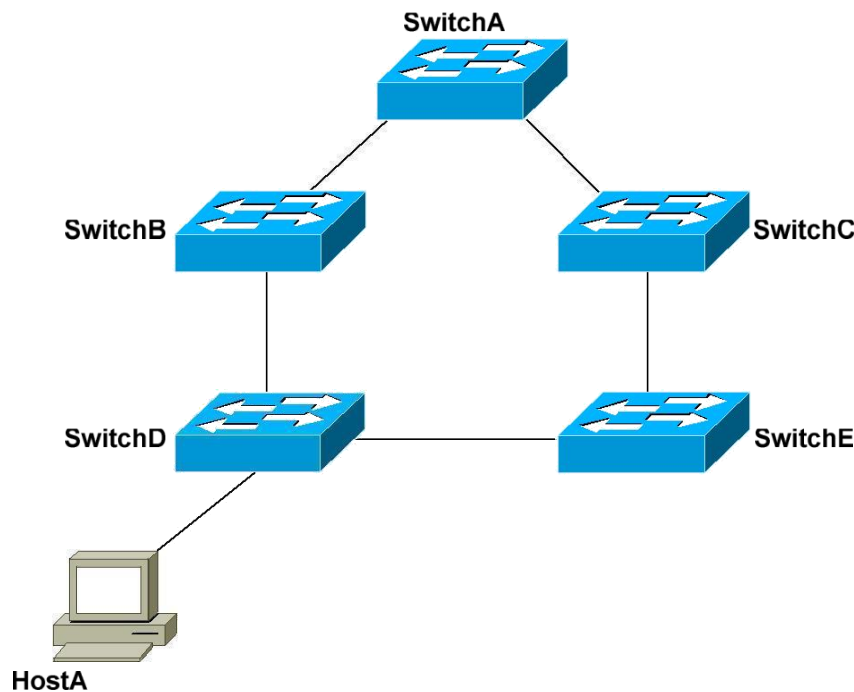
The Spanning Tree Protocol (STP) is a network protocol that builds a logical loop-free topology for Ethernet networks. The basic function of STP is to prevent bridge loops and the broadcast radiation that results from them. Spanning tree also allows a network design to include spare (redundant) links to provide an automatic backup path if an active link fails. This is done without the danger of bridge loops, or the need for manual enabling or disabling of these backup links. Select a root bridge. The *root bridge* of the spanning tree is the bridge with the smallest bridge ID. Each bridge has a configurable priority number and a MAC address; the bridge ID is the concatenation of the bridge priority and the MAC address. The bridge priority default is 32768 and can only be configured in multiples of 4096. When comparing two bridge IDs, the priority portions are compared first, and the MAC addresses are compared only if the priorities are equal. The switch with the lowest priority of all the switches will be the root; if there is a tie, then the switch with the lowest priority and lowest MAC address will be the root. For example, if switch A MAC=0200.0000.1111 and B MAC=0200.0000.2222 both have a priority of 32768 then switch A will be selected as the root bridge.^[c] If the network administrators would like switch B to become the root bridge, they must set its priority to be less than 32768.

Procedure:

Switching Loops

A Layer-2 switch belongs to only *one broadcast domain* and will forward both broadcasts and multicasts out *every* port but the originating port. When a **switching loop** is introduced into the network, a destructive **broadcast storm** will develop within *seconds*. A storm occurs when broadcasts are endlessly forwarded through the loop. Eventually, the storm will choke off all other network traffic.

Consider the following example:



If HostA sends out a broadcast, SwitchD will forward the broadcast out all ports in the same VLAN, including the trunk ports connecting to SwitchB and SwitchE. In turn, those two switches will forward that broadcast out all ports, including the trunks to the neighboring SwitchA and SwitchC.

The broadcast will loop around the switches infinitely. In fact, there will be two separate broadcast storms cycling in opposite directions through the switching loop. Only powering off the switches or physically removing the loop will stop the storm.

Spanning Tree Protocol (STP)

Spanning Tree Protocol (STP) was developed to prevent the broadcast storms caused by switching loops. STP was originally defined in **IEEE 802.1D**. Switches running STP will build a map or **topology** of the entire switching network. STP will identify if there are any loops, and then disable or *block* as many ports as necessary to eliminate all loops in the topology. A blocked port can be reactivated if another port goes down. This allows STP to maintain redundancy and fault-tolerance. However, because ports are blocked to eliminate loops, STP does not support load balancing unless an EtherChannel is used. EtherChannel is covered in detail in another guide.

STP switches exchange **Bridge Protocol Data Units (BPDU's)** to build the topology database. BPDU's are forwarded out all ports every **two seconds**, to a dedicated MAC multicast address of 0180.c200.0000.

Building the STP topology is a multistep **convergence** process:

- A **Root Bridge** is elected.
- **Root ports** are identified.
- Designated ports are identified.
- Ports are placed in a **blocking** state as required, to eliminate loops.

The **Root Bridge** serves as the central reference point for the STP topology. STP was originally developed when Layer-2 *bridges* were still prevalent, and thus the term *Root Bridge* is still used for nostalgic reasons. It is also acceptable to use the term **Root Switch**, though this is less common. Once the full topology is determined, and loops are eliminated, the switches are considered **converged**. STP is **enabled** by default on all Cisco switches, for all VLANs.

Electing an STP Root Bridge

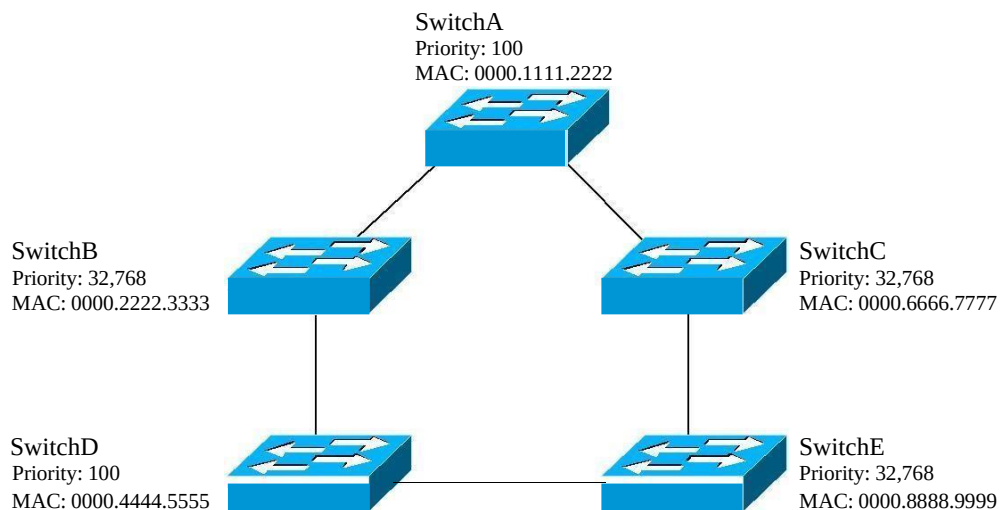
The first step in the STP convergence process is electing a **Root Bridge**, which is the central reference point for the STP topology. As a best practice, the Root Bridge should be the most centralized switch in the STP topology. A Root Bridge is elected based on its **Bridge ID**, comprised of two components in the original 802.1D standard:

16-bit **Bridge priority**

48-bit **MAC address**

The default priority is **32,768**, and the **lowest priority wins**. If there is a tie in priority, the **lowest MAC address** is used as the tie-breaker.

Consider the following example:



Switches exchange BPDU's to perform the election process, and the lowest Bridge ID determines the RootBridge:

- SwitchB, SwitchC, and SwitchE have the default priority of 32,768.
- SwitchA and SwitchD are tied with a lower priority of 100.
- SwitchA has the lowest MAC address and will be elected the Root Bridge.

By default, a switch will always believe it is the Root Bridge, until it receives a BPDU from a switch with a lower Bridge ID. This is referred to as a **superior BPDU**. The election process is continuous – if a new switch with the lowest Bridge ID is added to the topology, it will be elected as the Root Bridge.

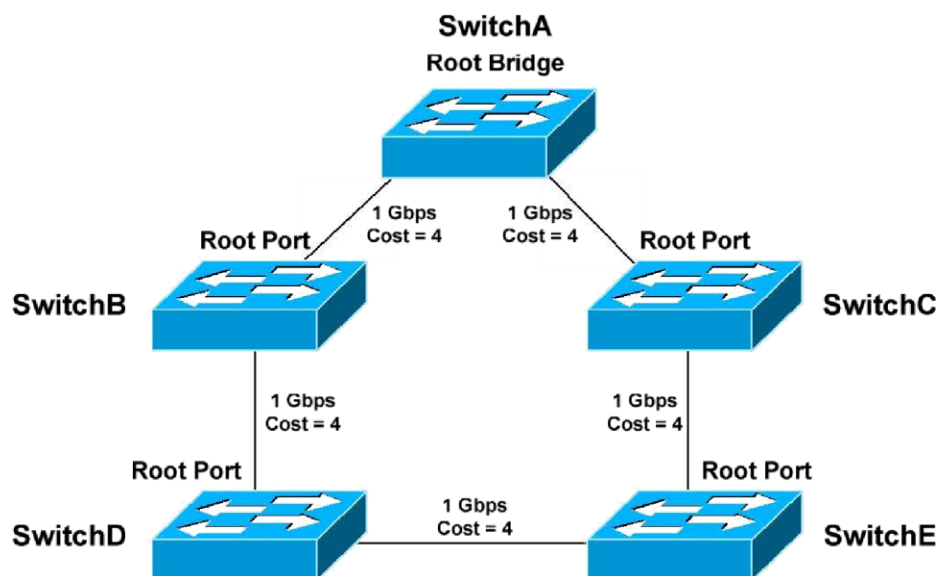
Identifying Root Ports

The second step in the STP convergence process is to identify **root ports**. The root port of each switch has the lowest **root path cost** to get to the Root Bridge. Each switch can only have *one* rootport. The Root Bridge *cannot* have a root port, as the purpose of a root port is to *point* to the Root Bridge.

Path cost is a *cumulative* cost to the Root Bridge, based on the bandwidth of the links. The *higher* the bandwidth, the *lower* the path cost:

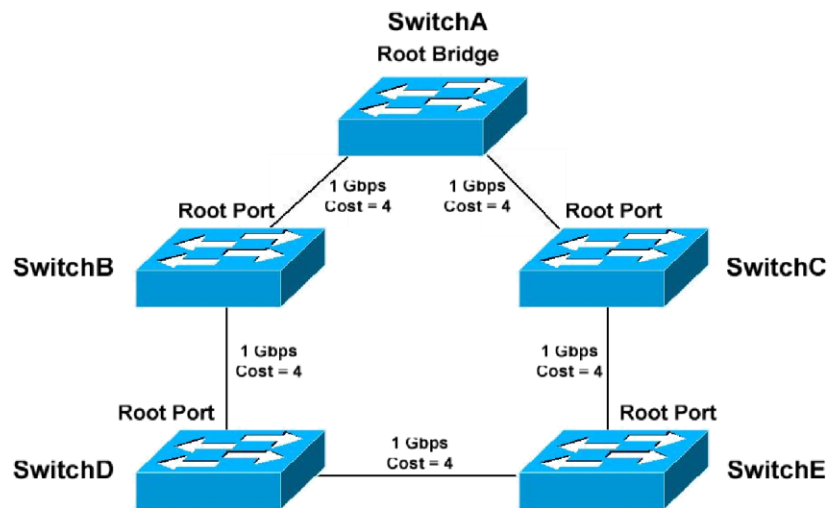
<i>Bandwidth</i>	<i>Cost</i>
10 Mbps	100
100 Mbps	19
10 Gbps	4
100 Gbps	2

A lower cost is preferred. Consider the following example:



Each 1Gbps link has a path cost of 4. SwitchA has a cumulative path cost of 0, because it is the Root Bridge. Thus, when SwitchA sends out BPDUs, it advertises a root path cost of 0. SwitchB has two paths to the Root Bridge:

- A direct connection to SwitchA, with a path cost of 4.
- Another path through SwitchD, with a path cost of 16.



The **lowest cumulative path cost** is considered **superior**, thus the port directly connecting to SwitchA will become the root port. A BPDU advertising a *higher* path cost is often referred to as an **inferior BPDU**.

SwitchD also has two paths to the Root Bridge:

- A path through SwitchB, with a path cost of 8.
- A path through SwitchE, with a path cost of 12.
- The port to SwitchB is preferred and will become the root port.

Recall that the Root Bridge will advertise BPDU's with a path cost of 0. As the downstream switches *receive* these BPDU's, they will add the path cost of the *receiving port*, and then advertise the cumulative cost to neighbors.

For example, SwitchC will receive a BPDU with a path cost of 0 from SwitchA, which is the Root Bridge. SwitchC will add the path cost of its receiving port, and thus SwitchC's cumulative path cost will be 4.

SwitchC will advertise a path cost of 4 to SwitchE, which will add the path cost of its receiving port. SwitchE's cumulative path cost will thus be 8.

Path cost can be artificially adjusted on a per-port basis:

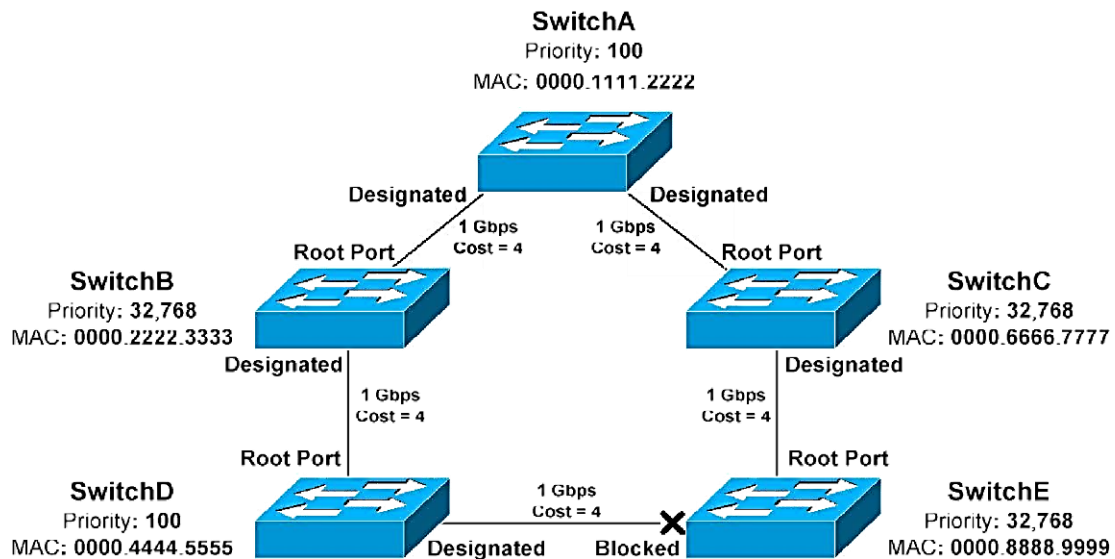
```
SwitchD(config)# int gi2/22
```

```
SwitchD(config-if)# spanning-tree vlan 101 cost 42
```

Identifying Designated Ports

The third step in the STP convergence process is to identify **designated ports**. A single designated port is identified for each network *segment*. This port is responsible for forwarding BPDUs and frames to that segment. If two ports are eligible to become the designated port, then there is a **loop**. One of the ports will be placed in a blocking state to eliminate the loop. Similar to a root port, the designated port is determined by the lowest cumulative path cost leading the Root Bridge. A designated port will *never* be placed in a blocking state unless there is a change to the switching topology and a more preferred designated port is elected.

Note: A port can never be both a designated port *and* a root port. Consider the following example:

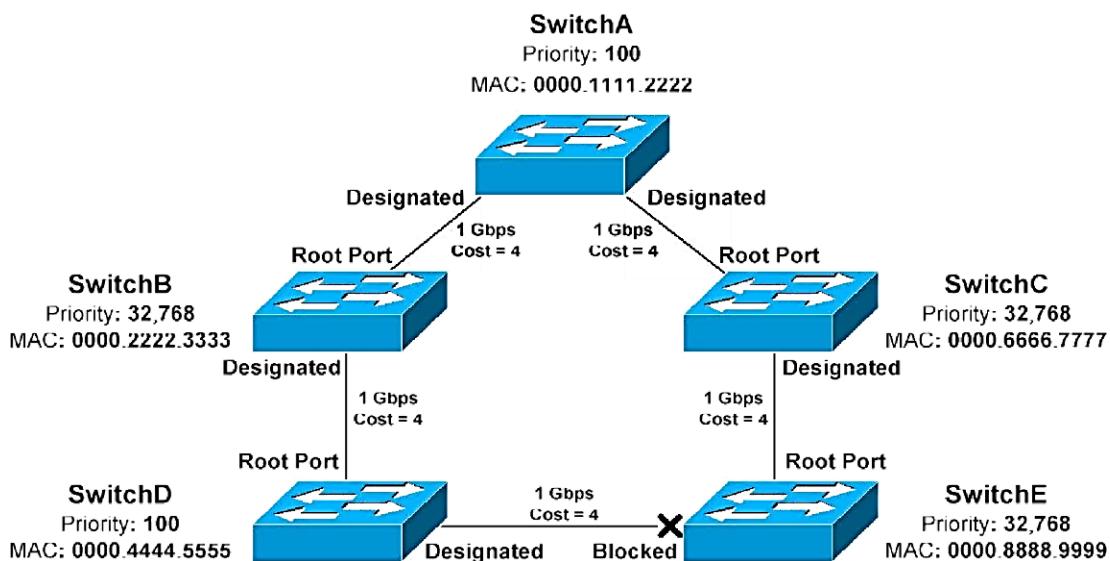


Ports on the Root Bridge are *never* placed in a blocking state. Thus, the two ports off of SwitchA will automatically become designated ports.

Remember, every network segment **must have one designated port**, regardless if a root port already exists on that segment.

Thus, the network segments between SwitchB and SwitchD, and between SwitchC and SwitchE, both require a designated port. The ports on SwitchD and Switch E have already been identified as root ports, thus the ports on Switch B and C will become the designated ports.

Note that the network segment between SwitchD and SwitchE does not contain a root port:



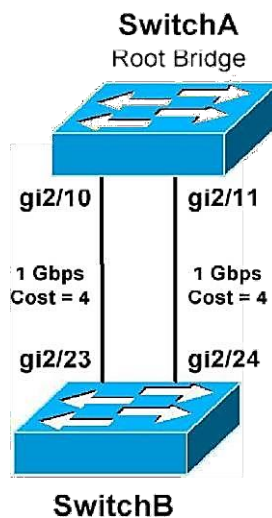
Because two ports on this segment are eligible to become the designated port, STP recognizes that a loop exists. One of the ports must be elected as the designated port, and the other must be placed in a blocking state. Normally, whichever switch has the lowest cumulative path cost will have its port become designated. The switch with the highest path cost will have its port blocked. In the

above example, there is a **tie** in cumulative path cost. Both SwitchD and SwitchE have a path cost of 12 to reach the Root Bridge on that segment. The **lowest Bridge ID** is used as the tiebreaker. SwitchD has a priority of 100, and SwitchE has the default priority of 32,768. Thus, the port on SwitchD will become the designated port. The port on SwitchE will be placed in a blocking state. As with electing the Root Bridge, if there is a tie in priority, the **lowest MAC address** is used as the tie breaker.

Remember: Any port not elected as a root or designated port will be placed in a blocking state.

Port ID

When electing root and designated ports, it is possible to have a tie in *both* path cost *and* Bridge ID. Consider the following example:



The bandwidth of both links is equal, thus both ports on SwitchB have an equal path cost to the Root Bridge. Which port will become the root port then? Normally, the lowest Bridge ID is used as the tiebreaker, but that is not possible in this circumstance.

Port ID is used as the final tiebreaker, and consists of two components:

- 4-bit **port priority**
- 12-bit **port number**, derived from the physical port number

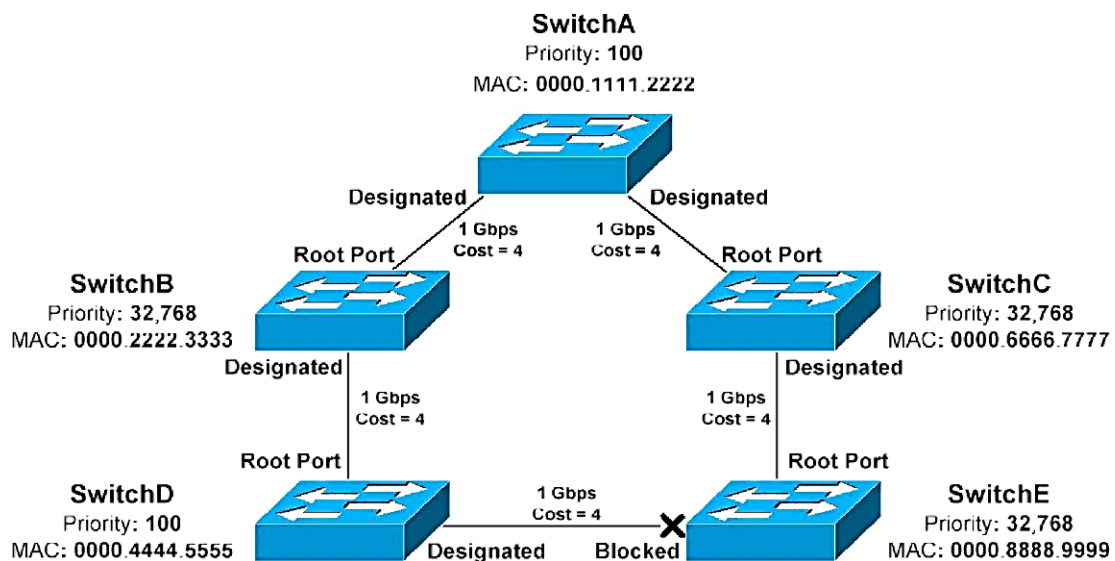
By default, the port priority of an interface is **128**, and a *lower* priority is preferred. If there is a tie in priority, the lowest port number is preferred.

The *sender* port ID determines the tie break, and not the *local* port ID. In the above example, SwitchB must decide whether *gi2/23* or *gi2/24* becomes the root port. SwitchB will observe BPDU's from SwitchA, which will contain the port ID's for *gi2/10* and *gi2/11*.

If priorities are equal, the sender Port ID from *gi2/10* is preferred, due to the lower port number. Thus, *gi2/23* on SwitchB will become the root port. The port number is a fixed value, but port priority can be changed on a per-interface basis:

```
Switch(config)# int gi2/11
```

```
Switch(config-if)# spanning-tree vlan 101 port-priority 32
```



Remember: Port ID is the *last* tiebreaker STP will consider. STP determines root and designated ports using the following criteria, *in order*:

- Lowest path cost to the Root Bridge
- Lowest bridge ID
- Lowest sender port ID

Note: Lowest Bridge ID is *always* used to determine the Root Bridge.

Basic STP Configuration

STP is **enabled** by default on all Cisco switches, for all VLANs and ports. PVST+ is the default STP mode on most modern Cisco platforms, allowing each VLAN to run a separate STP instance.

```
Switch>show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0006.2A11.24CC
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0006.2A11.24CC
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Desg FWD 19        128.1    P2p
Fa0/2        Desg FWD 19        128.2    P2p
```

```
Switch>show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID      Priority      32769
```

```
Address      0006.2A11.24CC
```

```
This bridge is the root
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority      32769 (priority 32768 sys-id-ext 1)
```

```
Address      0006.2A11.24CC
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Aging Time 20
```

```
Interface      Role Sts Cost      Prio.Nbr Type
```

```
-----
```

```
Fa0/1          Desg FWD 19      128.1    P2p
```

```
Fa0/2          Desg FWD 19      128.2    P2p
```

```
Switch>show spanning-tree summary
```

```
Switch is in pvst mode
```

```
Root bridge for: default
```

```
Extended system ID      is enabled
```

```
Portfast Default        is disabled
```

```
PortFast BPDU Guard Default is disabled
```

```
Portfast BPDU Filter Default is disabled
```

```
Loopguard Default       is disabled
```

```
EtherChannel misconfig guard is disabled
```

```
UplinkFast              is disabled
```

```
BackboneFast            is disabled
```

```
Configured Pathcost method used is short
```

```
Name      Blocking Listening Learning Forwarding STP Active
```

```
-----
```

```
VLAN0001      0          0          0          2          2
```

```
-----
```

```
1 vlans      0          0          0          2          2
```

The switch with the lowest Bridge ID is elected as the Root Bridge. The priority can be adjusted from its default of 32,768, to increase the likelihood that a switch is elected as the Root Bridge. Priority can be configured on a per-VLAN basis. Remember that the priority must be in multiples of 4,096 when extended system IDs are enabled. To change priority of a switch first you need to go to configuration mode to change the configuration of a switch then enter

Switch> enable

Switch# conf t

Switch(config)# spanning-tree vlan 1 priority 4096

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#spanning-tree vlan 1 priority 4096
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

After entering the above command exit from the configuration and then exit from the privilege mode and then enter below command to see whether the priority number of selected switch is changed or not

Switch> show spanning-tree active

It reports the effective priority value as 4097 (which is 1 higher than the configured value of 4096). This is because the switch adds 1 to the configured value to account for the root bridge's default priority.

```
Switch>show spanning-tree active
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0040.0B3D.0E38
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4097  (priority 4096 sys-id-ext 1)
             Address     0040.0B3D.0E38
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1 Desg FWD 19 128.1 P2p
Fa0/2 Desg FWD 19 128.2 P2p
```

STP can be disabled. This should be done with caution - any switching loop will result in a broadcast storm. To disable STP for an entire VLAN First you need to go to configuration mode:

Switch> enable

Switch# conf t

Switch(config)# no spanning-tree vlan 1

A range of VLANs can be specified:

Switch> enable

Switch# conf t

Switch(config)# no spanning-tree vlan 1 - 4094

STP can also be disabled on a per-port basis, for a specific VLAN:

Switch> enable

```
Switch# conf t
Switch(config)# interface gi2/23
Switch(config-if)# no spanning-tree vlan 101
Switch(config-if)# exit
Switch(config)# exit
Switch# exit
Switch>
```

A switch can be indirectly *forced* to become the Root Bridge for a specific VLAN:

```
SwitchA(config)# spanning-tree vlan 101 root primary
```

The *root primary* parameter automatically lowers the priority to 24,576. If another switch has a priority lower than 24,576, the priority will be lowered to 4,096 less than the current Root Bridge.

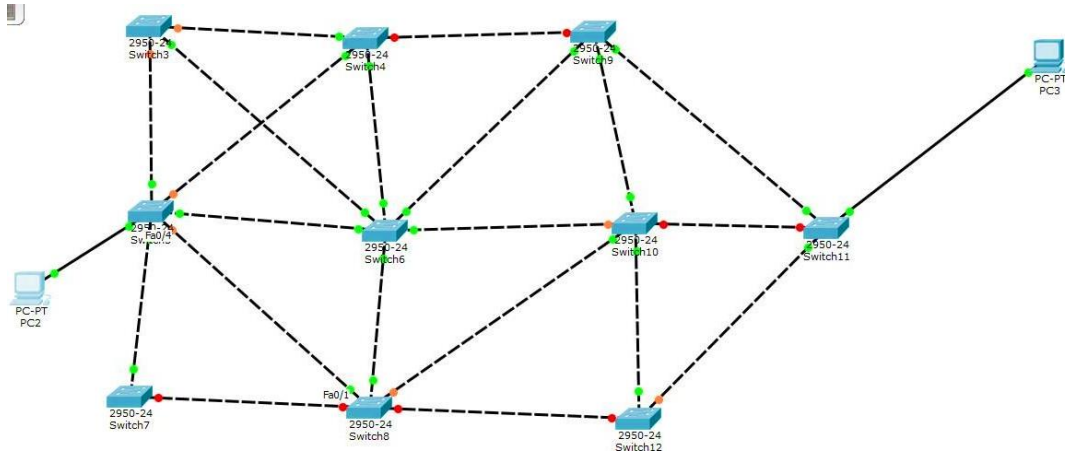
STP does not technically support a *backup* Root Bridge. However, the *root secondary* command can increase the likelihood that a specified switch will succeed as the new Root Bridge in the event of a failure:

```
SwitchB(config)# spanning-tree vlan 101 root secondary
```

The *root secondary* parameter in the above command automatically lowers the switch's priority to 28,672.

LAB TASK

a) **Simulate** a configuration of the network topology which contains two clients and ten switches find out which one is root among all switches, make one of them as a root, shut one link, and change priority of one switch. Design the given network.



Conclusion :
